

Cryptocurrency Financial Crime (2019–2025): Key Statistics

- **Total stolen by hacks (annual):** Crypto thefts surged from modest levels in 2019–20 to multi-billion sums later. 2020 saw $\approx \$0.52$ B stolen (global) ¹, jumping to $\$3.2$ B in 2021 ¹. 2022 was a record year at $\approx \$3.8$ B ²; 2023 dropped to $\approx \$1.7$ B ³ before rising to $\approx \$2.2$ B in 2024 ⁴. (Chainalysis estimates 2020 theft $\approx \$520$ M ¹, 2021 $\approx \$3.2$ B ¹, 2022 $\approx \$3.8$ B ², 2023 $\approx \$1.7$ B ³, 2024 $\approx \$2.2$ B ⁴.)
- **Ethereum-specific losses:** A large share of DeFi hacks occur on Ethereum. Estimated annual Ethereum ecosystem losses were $\approx \$415.7$ M (2020) ⁵, $\approx \$648.2$ M (2021) ⁶, $\approx \$528.9$ M (2022) ⁷, $\approx \$487.8$ M (2023) ⁸ and $\approx \$465.6$ M (2024) ⁹. (SlowMist “Hacked” reports; note 2019 shows no tracked Ethereum hacks ¹⁰.)

Year	Total Stolen (All Chains)	Stolen on Ethereum
2020	$\approx \$0.52$ B ¹	$\approx \$0.416$ B ⁵
2021	$\approx \$3.20$ B ¹	$\approx \$0.648$ B ⁶
2022	$\approx \$3.80$ B ²	$\approx \$0.529$ B ⁷
2023	$\approx \$1.70$ B ³	$\approx \$0.488$ B ⁸
2024	$\approx \$2.20$ B ⁴	$\approx \$0.466$ B ⁹
Total (2019–24)	$\approx \\$8.92$ B	$\approx \\$2.547$ B

- **Breakdown by platform (DeFi vs. CEX):** The lion’s share of funds has come from DeFi exploits. In 2021 and 2022, ~ 72 – 82% of stolen crypto was from DeFi protocols ¹ ¹¹. For example, Chainalysis reports $\$2.3$ B of $\$3.2$ B stolen in 2021 (72%) was from DeFi ¹; TRM similarly finds $\sim \$3.0$ B of $\$3.7$ B stolen in 2022 ($\sim 80\%$) involved DeFi targets ¹¹. In 2023 this trend eased: $\sim 63\%$ of the $\$1.7$ B stolen was from DeFi ($\$1.1$ B) ¹², with the rest ($\sim \$0.6$ B) from centralized exchanges (CEX) and other off-chain sources. Table below summarizes annual DeFi vs. CEX losses (estimates):

Year	DeFi Exploits ($\approx$ USD) 【source】	CEX Exploits ($\approx$ USD)
2020	$\approx \$0.162$ B ¹³	$\approx \$0.358$ B
2021	$\approx \$2.30$ B ¹	$\approx \$0.90$ B
2022	$\approx \$3.00$ B ¹¹	$\approx \$0.80$ B
2023	$\approx \$1.10$ B ¹²	$\approx \$0.60$ B

(Values inferred: “CEX” = total – DeFi. 2020 totals from ¹³ ¹ ; 2021–22 from ¹ ¹¹ ; 2023 from ¹² .)

- **Recoveries/Seizures:** A small portion of stolen crypto is recovered. Chainalysis notes ~\$12.6 B in illicit crypto has been frozen or seized globally (all crime types) as of early 2025 ¹⁴ . Notable law-enforcement recoveries include ~\$3.36 B seized from the 2012 Silk Road hacker ¹⁵ and ~\$1.0 B seized from Silk Road exchange funds ¹⁶ . In 2021 the U.S. IRS-CI reported \$3.5 B seized (93% of its seizures) ¹⁷ (mostly crypto). Despite these actions, only a minority of stolen assets are ultimately returned to victims.
- **Laundering via bridges & mixers:** Criminals increasingly use bridges and mixers to obfuscate stolen crypto. Chainalysis reports funds sent to mixers were ~\$1.0 B in 2022 and \$0.504 B in 2023 ¹⁸ . Bridge usage jumped: illicit funds to cross-chain bridges totaled \$0.312 B in 2022 and \$0.744 B in 2023 ¹⁹ . (By contrast, in 2021 mixers processed ~\$11.5 B total but only ~\$1.15 B illicit ²⁰ .) In total, ~\$1.504 B has been laundered via mixers and ~\$1.056 B via bridges in 2022–23.

Year	Funds to Mixers (USD) 【source】	Funds to Cross-Chain Bridges (USD) 【source】
2022	~\$1.00 B ¹⁸	~\$0.312 B ¹⁹
2023	~\$0.504 B ¹⁸	~\$0.744 B ¹⁹
Total	~\$1.504 B	~\$1.056 B

- **Phishing scams:** Crypto phishing (fake sites/app approvals, impersonation, etc.) remains costly. Chainalysis analysis of *approval-phishing* scams (tricking victims to approve token transfers) found an estimated \$516.8 M stolen in 2022 and ~\$374.6 M through November 2023 ²¹ . (Chainalysis tracked ~1,013 addresses involved in such targeted phishing since 2021 ²² , but total incident counts are vastly higher and not fully quantified.) In sum, crypto scam losses (including phishing) ran into the billions annually at peaks ²¹ .
- **Common vulnerabilities (DeFi & CEX):** Hackers exploit both on-chain code flaws and off-chain lapses. **DeFi exploits:** *Smart-contract bugs* dominate. The most frequent flaw is missing/faulty input validation (e.g. integer overflows) – cited in ~34.6% of contract hacks ²³ – and Chainalysis/TRM report that ~90% of 2022 DeFi hack losses stemmed from poor authentication/validation logic ²⁴ . *Reentrancy bugs* remain notable (resurfacing strongly in 2023 ²⁵). *Flash-loan attacks* surged (comprising 83.3% of DeFi exploits in 2024 ²⁶) by allowing rapid market manipulation. Other contract attack vectors include oracle/price manipulation and governance exploit (e.g. malicious admin keys). **Off-chain/CEX exploits:** The single largest factor is compromised keys/credentials: Halborn reports *compromised accounts* (leaked admin or private keys) caused over 50% of crypto theft incidents ²⁷ . Many major exchange hacks (e.g. KuCoin 2020) were due to hot-wallet key theft or insider misuse. Social-engineering and stolen API keys also figure. Overall, vulnerabilities rank roughly: (1) account/key compromise, (2) smart-contract auth/validation bugs, (3) reentrancy & logic bugs, (4) flash-loan/oracle attacks, (5) governance flaws, etc. ²⁴ ²³ ²⁵ ²⁶ .

Sources: Chainalysis (Crypto Crime Reports, blogs) ¹ ³ ⁴ ¹⁴ ¹⁵ ¹⁸ ¹⁹ ²¹ ; TRM Labs (2023 report) ¹¹ ²⁴ ; SlowMist hack statistics ⁵ ⁷ ⁶ ⁸ ⁹ ; Halborn Top 100 DeFi Hacks ²³ ²⁵ ²⁶ .

1 13 17 **Crypto Crime Trends for 2022: Illicit Transaction Activity Reaches All-Time High in Value, All-Time Low in Share of All Cryptocurrency Activity - Chainalysis**

<https://www.chainalysis.com/blog/2022-crypto-crime-report-introduction/>

2 20 **pensamientopenal.com.ar**

https://www.pensamientopenal.com.ar/system/files/Documento_Editado1090_0.pdf

3 12 **Stolen Crypto Falls in 2023, but Hacking Remains a Threat**

<https://www.chainalysis.com/blog/crypto-hacking-stolen-funds-2024/>

4 **\$2.2 Billion Stolen in Crypto in 2024 but Hacked Volumes Stagnate**

<https://www.chainalysis.com/blog/crypto-hacking-stolen-funds-2025/>

5 **SlowMist Hacked - SlowMist Zone**

<https://hacked.slowmist.io/statistics/?c=ETH&d=2020>

6 **SlowMist Hacked - SlowMist Zone**

<https://hacked.slowmist.io/statistics/?c=ETH&d=2021>

7 **SlowMist Hacked - SlowMist Zone**

<https://hacked.slowmist.io/statistics/?c=ETH&d=2022>

8 **SlowMist Hacked - SlowMist Zone**

<https://hacked.slowmist.io/statistics/?c=ETH&d=2023>

9 **SlowMist Hacked - SlowMist Zone**

<https://hacked.slowmist.io/statistics/?c=ETH&d=2024>

10 **SlowMist Hacked - SlowMist Zone**

<https://hacked.slowmist.io/statistics/?c=ETH&d=2019>

11 24 **Illicit Crypto Ecosystem Report**

<https://www.trmlabs.com/resources/reports/the-illicit-crypto-ecosystem-report-2022>

14 15 16 **Asset Seizure and Cryptocurrency - Chainalysis**

<https://www.chainalysis.com/blog/cryptocurrency-asset-seizure/>

18 19 **2024 Crypto Money Laundering Report - Chainalysis**

<https://www.chainalysis.com/blog/2024-crypto-money-laundering/>

21 22 **Crypto Crime: Targeted Approval Phishing Scams On the Rise**

<https://www.chainalysis.com/blog/approval-phishing-cryptocurrency-scams-2023/>

23 25 26 27 **The Top 100 DeFi Hacks Report 2025**

<https://www.halborn.com/reports/top-100-defi-hacks-2025>